YUCHEN DU

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SUMMARY

Ph.D. candidate at Purdue University and former **Amazon Applied Scientist Intern** building **reliable LLM agents** for online decision-making. Current research evaluates tool-using agents that balance service quality and operating efficiency in dynamic fleet assignment; at Amazon, developed and deployed a **two-stage agent** that helped investigators adjudicate fraud-related seller appeals using multimodal evidence, verification actions, and historical human corrections. Published research spans machine learning, stochastic optimization, and simulation.

SKILLS

Applied Machine Learning: Classification, Natural Language Processing, Statistical Analysis, Transformers, BERT, LSTM/GRU, Topic Modeling

LLM & Retrieval Systems: RAG, GraphRAG, Knowledge Graphs, Vector Retrieval, Structured Outputs, Schema Validation

Data & Evaluation: Human-in-the-Loop Review, Controlled Ablations, Error Analysis, Accuracy, Precision/Recall

Programming & Systems: Python, SQL, PyTorch, Scikit-learn, LangGraph, Amazon Bedrock, Gurobi

EDUCATION

Purdue University, Ph.D. in Industrial Engineering Technology	2023-Expected Dec 2026
University of Michigan, MSc in Data Science track at Industrial and Operations Engineering	2021-2023
South China University of Technology, BSc in Industrial Engineering	2017-2021

EXPERIENCE

AMAZON | APPLIED SCIENTIST INTERN — Seattle, WA | Aug 2025 - Nov 2025

Selling Partner Trust & Store Integrity (TSI) - Fraud Detection Team

- Built and deployed an AI decision-support system for appeals from sellers whose accounts had been restricted by **fraud-detection controls**, helping investigators synthesize fragmented evidence, distinguish false-positive restrictions from valid enforcement, and decide whether to reinstate or keep accounts blocked.
- Developed a two-stage LLM agent on Amazon Bedrock using RAG with **graph-based memory**: offline, it transformed multi-turn seller-review records, verification actions, and multimodal evidence into Evidence-Action-Factor-Decision (EAFD) graphs; online, GraphRAG retrieved similar cases to produce evidence-grounded recommendations.
- When evidence was insufficient, the agent returned a Request More Information outcome with targeted verification steps and missing evidence; across 260+ cases, it improved offline alignment from 70.8% to 95.8%, reached 96.3% cumulative production alignment, and maintained **100% precision on approvals**.

SELECTED PAPERS

[1] **Yuchen Du**, Ashley Li, and Zixi Huang. Reviewing the Reviewer: Graph-Enhanced LLMs for E-commerce Appeal Adjudication. *arXiv preprint arXiv:2603.19267*, 2026. [arXiv]

[2] **Yuchen Du** and Tho V. Le. Understanding Social Perceptions, Interactions, and Safety Aspects of Sidewalk Delivery Robots Using Sentiment Analysis. *Transportation Research Record*, article 03611981251394686, 2025. [DOI]

- Built an end-to-end ML/NLP pipeline on 10K+ comments, comparing SVM and BERT+LSTM+GRU classifiers; achieved 0.78 three-class accuracy and used LDA topic modeling to identify safety and adoption concerns.

[3] **Yuchen Du** et al. Evaluating Bounded Language-Model Interventions for Online Assignment in Dynamic Pickup-and-Delivery. Under review at *AAAI 2027*, 2026.

- Built a tool-using LLM agent for online fleet assignment: it read requests, robot locations and routes, capacity, and time windows through diagnostic tools, then balanced delivery service, operating efficiency, and future fleet capacity with deterministic verification before applying each assignment to the fleet.
- Across 200 cases with varied demand patterns, the agent cut window lateness by 88.03% versus the strongest action-matched heuristic at a 2.86% operating-time premium, with the aggregate objective nearly unchanged (-0.089%).

[4] **Yuchen Du**, Hai Yang, Joseph Y. J. Chow, and Tho V. Le. Two-stage stochastic fleet and battery sizing with routing optimization for sidewalk delivery robots. *Transportation Research Part E*, 201:104220, 2025. [DOI]

[5] Hai Yang, **Yuchen Du**, Tho V. Le, and Joseph Y. J. Chow. Analytical model for large-scale design of sidewalk delivery robot systems with bounded makespans. *Transportation Research Part C*, 170:104978, 2025. [DOI]